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AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science

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Edited by Timothy D. Wilson, University of Virginia, Charlottesville, VA; received September 22, 2025; accepted March 16, 2026

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Large Language Models (LLMs) such as ChatGPT are transforming how scientists conduct and validate research, offering promise as tools to improve scientific reproducibility. However, computational reproducibility and error detection remain expensive and labor-intensive. We experimentally test how collaboration between researchers and LLM assistants influences the reproduction of quantitative social science findings across different levels of AI autonomy. We randomly assigned 288 researchers to 103 teams working under three conditions: human-only, AI-assisted (using ChatGPT as a collaborative tool), or AI-led (ChatGPT operating with minimal human oversight). Teams reproduced published results from leading social science journals, detected coding errors, and proposed robustness checks. Human-only and AI-assisted teams achieved comparable reproduction rates (94% vs. 91%) and performed similarly on most outcomes, except human-only teams identified significantly more major coding errors. Both substantially outperformed AI-led teams, which achieved only a 37% reproduction rate, detected fewer errors across all categories, proposed weaker robustness checks, and required more time. This autonomous approach, however, likely represents only a lower bound of AI capabilities. Despite rapid model advances, expert human judgment currently remains indispensable for reliable empirical verification. While AI assistance did not degrade most outcomes, it provided no measurable advantages and was associated with reduced detection of major errors. However, the 37% autonomous reproduction rate indicates that AI could provide value in settings where scale or cost constraints preclude human review of papers, even though general-purpose LLMs offer no immediate advantages for human-supervised verification.

AI | reproducibility | large language models

Reproducibility is a cornerstone of robust quantitative empirical research, where complex methodologies and data handling techniques are common (1–8). Despite advancements in reproducibility protocols (9), concerns persist regarding the accuracy and reliability of published findings (10–17). Unclear reporting and methodological advances requiring expertise when evaluating quantitative studies contribute to the current reproducibility and replication crises in the behavioral and social sciences. At the same time, verifying computational reproducibility remains costly and labor-intensive (18). Even when journals require replication packages, reproducing results often involves navigating complex scripts, large datasets, and intricate empirical workflows. As empirical research becomes increasingly complex, scalable approaches to verification are needed to ensure that the reliability of published findings can be efficiently assessed.

This study investigates how artificial intelligence (AI) tools, such as Large Language Models (LLMs), could support researchers, data editors, and scientific journals in computationally reproducing research. We focus on three modes of AI and human interaction: human-only teams, human teams with AI assistance (the "AI-assisted" approach), and teams that provided only limited oversight while AI carried out reproducibility checks (the "AI-led" approach). The AI-led approach approximates a "protoagentic" system: an LLM tasked with reasoning through a reproducibility exercise with minimal human supervision. We use ChatGPT because it processes different file formats effectively for reproduction and is used most frequently by researchers (19).

This paper tests how effectively AI supports reproduction of scientific articles and works in complex cases where coding errors or methodological inconsistencies arise. We employ a randomized controlled trial design involving three treatment arms. We contribute to a large literature documenting the benefits and limitations of human–AI integration, as well as full automation (20). Evidence from human–AI decision-making suggests that performance ordering between AI alone, human alone, and human–AI teams is mixed and task-dependent, and that human–AI combinations often fail to outperform AI alone, sometimes performing worse due to miscalibrated trust and under- or overreliance on AI assistance (21–34). This is crucial for science because current methods for performing computational reproducibility and robustness checks are expensive, time consuming, and require advanced technical skills (18, 35). We also contribute to a growing body of literature documenting the potential pitfalls of integrating human and artificial intelligence, such as overreliance and expertise erosion (36, 37). This research also provides some comparative productivity measures in highly specialized intellectual tasks. This line of research mainly focuses on customer support agents and low-skill occupations, whereas we study high-skill scientific reproducibility tasks (29, 38).

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Competing interest statement: The views expressed in this paper are those of the authors. No responsibility for them should be attributed to the Bank of Canada. A. Marcoci is a UK Research and Innovation Policy Fellow seconded to the Department for Science, Innovation and Technology. The views and conclusions contained herein are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the Department for Science, Innovation and Technology or the UK Government.

This article is a PNAS Direct Submission.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2524747123/-/DCSupplemental.

Published May 28, 2026.

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We focus on three groups of outcomes across the treatment arms: 1) computational reproducibility (success rate and time required), 2) error detection capabilities, and 3) proposing and implementing quality robustness checks. Understanding the impact of the treatment on these outcomes contributes to a broader understanding of AI, and offers insights into the optimal balance of human and AI involvement in research tasks.

1. Procedures

The first 10 coauthors organized seven AI replication games between February and November 2024, including a pilot in February. All remaining coauthors and a few of the organizers participated in one of those games. The participating coauthors were a mix of master and PhD students, postdoctoral fellows, professors, and researchers from nonacademic organizations with a doctoral degree. *SI Appendix*, Table S8 provides details on team composition. Randomization was carried out in two steps for each of the seven events. In step one, participants were randomly assigned to a team of three to evaluate the reproducibility of a quantitative social science article. The randomization in step one was conditional on the software preferences reported by participants (Stata or R) and the mode of participation (in person or virtual). In step two, each team was randomly assigned to one of three treatment arms: human-only, AI-assisted, or AI-led.

Each team was assigned a study from leading social science journals (i.e., economics, political science, or behavioral science/psychology). Each event included two studies with known coding errors (one in Stata and one in R) that had been identified by the lead authors in a prior study but were not publicly disclosed at the time of the AI replication game. Detailed information about the papers used that contained coding errors can be found in *SI Appendix*, Tables S1 and S2. Descriptions of the coding errors identified prior to each replication game can be found in *SI Appendix*, Table S3 through *SI Appendix*, Table S4. Coding errors occurred when preparing data for analysis (variable definitions, incorrect merging of datasets, differing sample restrictions, not cleaning variables, missing variables) as well as when carrying out the analysis (discrepancies between code and what is written in the article). Examples of the latter include inconsistently specified SEs and control variables. Teams and local organizers had no information about the study they would be reproducing until the start of the event. Twelve studies were used in total, with a few reused for multiple events.

Relevant resources were given to the teams at 09:00 local time on the day of the event. We shared with them: the journal article and online appendix as PDFs, the original authors' replication package, and screenshots of the exhibit to reproduce from the article (*SI Appendix*). Screenshots were introduced after the pilot event to assist AI-led teams, as the AI might be better able to process tables and figures as images rather than when embedded in PDF files. Teams had seven hours to complete three tasks: i) computationally reproduce a few predetermined results, ii) detect coding errors, and iii) suggest and implement up to two robustness checks. The three tasks were independent from each other (e.g., teams did not need to fix coding errors to computationally reproduce results). However, teams were instructed to begin with reproducing the results before proceeding to specifically search for coding errors and propose robustness checks. Teams could leave before the end of the event if they believed they had completed their tasks as feasibly as possible. Upon completion, teams were asked to email the lead authors a (templated) time log documenting whether they completed computational reproducibility, with a list of all coding errors uncovered, and two robustness checks. All AI-assisted and AI-led teams used ChatGPT during the event and had to provide their AI conversation history (i.e., a transcript of all prompts and responses exchanged with ChatGPT).

Participants were offered coauthorship on this paper, independent of their team's performance or success in reproducing results. No monetary compensation or performance-based incentives were provided. While this may have led to reduced effort for some teams, it also reduced incentives for strategic behavior or protocol violations, particularly for AI-led teams who were asked not to directly examine the article, code, or data.

Access to a paid subscription of ChatGPT (powered initially by GPT-4 and subsequently by other models) was provided to all members in the AI-assisted and AI-led teams. While ChatGPT had six different versions available between February 14th 2024 (training for our pilot) and our final event on November 22nd 2024, researchers had access to the main flagship models (GPT-4, and/or GPT-4o). These models were capable of processing files, equipped with a Python environment for interpreting code and conducting data analysis, and had internet access. Additional information on the

Significance

Verifying results of published social sciences research is essential but expensive, costing hundreds of dollars per study. With AI tools like ChatGPT becoming widespread, we tested whether they could help scientists check if research findings can be reproduced. We assigned 288 researchers to 103 teams working with no AI, with AI as an assistant, or AI leading the work with minimal human input. Human teams and AI-assisted teams performed similarly on most tasks, but humans caught more critical errors. AI working autonomously achieved a 37% reproduction rate, making it potentially useful for automated screening when human review is cost-prohibitive. These results nonetheless show that human expertise remains essential for reliable scientific validation.

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different versions and use by teams at events are included in *SI Appendix*, ChatGPT Models.

AI-assisted and AI-led teams took part in a mandatory one-hour training on the usage of ChatGPT. Participants viewed the training live or later via recording. The AI training was optional for human-only teams. The training had nine components which we outline here but describe further in *SI Appendix*, AI Training: 1) Introduction, Overview of ChatGPT and Access; 2) Interaction with ChatGPT; 3) Sharing Chats with I4R; 4) Coding Assistance; 5) Uploading Files and Images; 6) Conducting Data Analysis Using ChatGPT; 7) ChatGPT API; 8) Customizing ChatGPT; and 9) Explanation of Differences Among ChatGPT Models. AI-led and AI-assisted teams constructed their own prompts but were given examples and best-practice guidance in the training session. Using textual analysis on all prompts, we show limited overlap in prompt wording across AI-led and AI-assisted teams (*SI Appendix*).

The human-only teams were not allowed to use ChatGPT or any other AI tool. The AI-assisted teams were allowed to use ChatGPT without limitation (but no other AI tool). AI-led teams had to perform the tasks only using the guidance of ChatGPT. They were not allowed to read the article or look at the data and code but could ask ChatGPT to summarize the article. They had to upload the article to ChatGPT along with an image of the table(s)/figure(s) to be reproduced, the replication code, and the data files where feasible. They were asked to first use ChatGPT's Python interpreter module to conduct the analysis. However, they were allowed to run analysis code locally (in R or Stata) when ChatGPT failed to run the analysis itself. When running code locally, the teams were not allowed to use any other code except the one provided by ChatGPT, though the teams could adjust file paths and their environment without the assistance of ChatGPT. During the pregames AI training, participants were shown examples of how to upload the article and replication files to ChatGPT and how to use the Python interpreter module. We relied on the integrity of the AI-led teams to not look at the studies, code, or files. That is, we asked them to pass everything through ChatGPT; we did not give specific guidance on how teams should operate. Teammates could work independently or jointly throughout the event.

In summary, we have 103 teams: 33 human-only teams (92 researchers), 35 AI-assisted teams (93 researchers), and 35 AI-led teams (103 researchers). *SI Appendix*, Table S8 shows the treatment arms are balanced across observables.

1.1. Three Tasks. Participants had three objective tasks with measurable outcomes. First, teams were asked to computationally reproduce a few selected results in the study assigned to them. The numerical results were selected by the lead author, AB, based on their relative importance to the main claims of the article. Computational reproducibility involves using the same data as the original authors and running their code. In the templated log, teams recorded the time taken to computationally reproduce the numerical result. Notably, AB, JA, and DM were able to computationally reproduce the selected results before the event, requiring only minimal adjustments (e.g., updating file paths). We have two different dependent variables for computational reproducibility: one outcome as a binary variable (completed computational reproducibility vs. did not complete), and one that is time (in minutes) from the start of the event to when teams completed a computational reproduction. A computational reproduction is defined as the successful execution of the original authors' codes and the production of numerical results in line with those in the article.

Second, we compare how effective different team types were in finding coding errors or data irregularities. For simplicity, we refer to these as "errors." We categorize errors as major or minor based on whether they could, in theory, have an impact on the claims tested. For instance, a coding error or data irregularity that impacts the dependent or independent variables is considered a major error, as it could have an impact on the estimation results. In contrast, minor coding errors are typically easily fixed by the reproducers and do not impact the validity of the claims made by the original authors. In a set of exploratory analyses, we also categorize coding errors along three dimensions: i) whether the error occurs in preparing the data and analysis, ii) whether the error is related to the regression analysis, and iii) whether it is a transcription error (e.g., a mismatch between the coefficient reported in the article and the coefficient produced by the code, such as −0.034 vs. 0.034). We also investigate the extent of false error detection and the share of errors not uncovered by the treatment arm.

Third, we asked each team to propose and perform two robustness checks. A robustness check is defined in our study as an additional statistical computation. We instructed that these robustness checks should not repeat ones already mentioned in the study or its *SI Appendix*, that they should be feasible, and that heterogeneity analysis (e.g., comparing female and male respondents) was not considered a robustness check.

Defining what makes a robustness check "good" or "bad" is not straightforward. We define four binary criteria for evaluating the quality of robustness checks: i) clarity of purpose and execution; ii) feasibility; iii) novelty (i.e., not previously done by the original authors); and iv) relevance to the validity of the empirical strategy. Items i) through iii) are basic necessary conditions. Item iv) requires that the purpose of the robustness test is to provide evidence regarding the credibility of the empirical strategy (39–41). All four criteria must be met for a robustness check to be considered "good." Additionally, running corrected code in an attempt to correct major errors in the original paper is coded as a "good" robustness check, regardless of whether it complies with the previous criteria.

We measure differences by team type in proposing and implementing robustness tests using four measures. The first two are whether teams proposed one or two "good" robustness checks. The third and fourth dependent variables are whether the participants report to have implemented one or two of those "good" robustness checks, respectively.

2. Results

Our analyses were preregistered after the pilot event in Toronto. We list deviations from our preregistration in *SI Appendix* and note throughout whether the analysis is exploratory.

2.1. Computational Reproducibility. Our main finding is that computational reproducibility rates varied substantially across the groups. Most human-only (94%; 31/33) and AI-assisted (91%; 32/35) teams could computationally reproduce the results, while only 37% (13/35) of AI-led teams were able to do so (Table 1). Table 2 shows the ordinary least squares (OLS) estimates of our main regression model (see *SI Appendix*, Table S10 for logit and Poisson regressions and *SI Appendix*, Table S12 for coefficient estimates concerning the control variables). We find that human-only teams are about 59 percentage points more likely than

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Table 1. Comparison of human, AI-assisted, and AI-led metrics

Variable	Human-only	AI-assisted	AI-led	Human-only vs. AI-assisted	Human-only vs. AI-led	AI-assisted vs. AI-led
Reproduction	0.939 (0.242)	0.914 (0.284)	0.371 (0.490)	0.025 [0.697]	0.568 [<0.001]	0.543 [<0.001]
Minutes to reproduction	82.0 (39.8)	93.3 (85.4)	179.7 (68.4)	-11.3 [0.505]	-97.7 [<0.001]	-86.4 [0.002]
Number of minor errors	1.000 (1.658)	1.400 (2.488)	0.686 (1.605)	-0.400 [0.441]	0.314 [0.430]	0.714 [0.158]
Minutes to first minor error	141.6 (97.0)	139.9 (83.1)	157.6 (85.6)	1.7 [0.960]	-16.0 [0.691]	-17.7 [0.622]
Number of major errors	1.697 (2.568)	0.743 (1.120)	0.229 (0.547)	0.954 [0.049]	1.468 [0.002]	0.514 [0.017]
Minutes to first major error	110.5 (69.5)	130.3 (86.9)	152.8 (94.3)	-19.8 [0.487]	-42.3 [0.261]	-22.5 [0.606]
At least one good robustness check	1.000 (0.000)	1.000 (0.000)	0.829 (0.382)	0.000 [NA]	0.171 [0.012]	0.171 [0.010]
At least two good robustness checks	0.879 (0.331)	0.857 (0.355)	0.629 (0.490)	0.022 [0.796]	0.250 [0.017]	0.229 [0.029]
Ran at least one good robustness check	0.939 (0.242)	0.943 (0.236)	0.571 (0.502)	-0.003 [0.953]	0.368 [<0.001]	0.371 [<0.001]
Ran at least two good robustness checks	0.788 (0.415)	0.800 (0.406)	0.457 (0.505)	-0.012 [0.903]	0.331 [0.005]	0.343 [0.003]

Note: Columns 2–4 present means and SEs in parentheses for individual groups (Human-only, AI-Assisted, and AI-Led); columns 5–7 present differences in means and *P*-values in brackets for group comparisons (Human-Only vs. AI-Assisted, Human-Only vs. AI-Led, and AI-Assisted vs. AI-Led).

AI-led teams to successfully computationally reproduce the results ($P < 0.001$). In contrast, there is no statistically significant difference between human-only and AI-assisted teams ($P = 0.771$).

We next investigate how the distribution of time-to-computational reproduction varies across groups. Fig. 1 plots complementary Kaplan–Meier curves showing, by treatment arm, how long teams took to reproduce their paper by the end of the event. The proportion of teams that reproduce their paper does not reach 100% after seven hours in any treatment arm because all treatment arms contain some teams who could not reproduce their paper. This is especially noticeable for AI-led teams. We find that human-only and AI-assisted teams are significantly faster than AI-led teams (Table 1). There is no statistically significant difference between human and AI-assisted teams.

In an exploratory analysis, we investigate whether AI-assisted and AI-led teams improved over time. In our setting, improvements could be due to new ChatGPT versions and increased researchers' skills over time. In *SI Appendix*, Fig. S2, we show the difference in computational reproducibility rates between the treatment groups by event. Visually, AI-led teams did not improve over time when compared to human-only teams during the first five events in 2024. We observe that the reproducibility

Table 2. Causal relationship between treatment groups and reproducibility outcomes

	(1) Reproduction	(2) Minor errors	(3) Major errors	(4) One good robustness	(5) Two good robustness	(6) Ran one robustness	(7) Ran two robustness
AI-assisted	-0.018	0.313	-1.022***	-0.009	-0.014	-0.032	-0.009
	(0.063)	(0.387)	(0.362)	(0.027)	(0.103)	(0.061)	(0.113)
	[-0.144; 0.107]	[-0.458; 1.083]	[-1.743; -0.300]	[-0.063; 0.046]	[-0.220; 0.191]	[-0.155; 0.090]	[-0.233; 0.216]
AI−led	-0.593***	-0.331	-1.344***	-0.167**	-0.250**	-0.323***	-0.290**
	(0.090)	(0.350)	(0.342)	(0.068)	(0.107)	(0.098)	(0.126)
	[-0.773; -0.413]	[-1.029; 0.366]	[-2.024; -0.664]	[-0.302; -0.031]	[-0.463; -0.037]	[-0.518; -0.127]	[-0.540; -0.040]
Controls	✓	✓	✓	✓	✓	✓	✓
Mean of dep. var	0.738	1.029	0.874	0.942	0.786	0.816	0.680
P-val (AI-assisted = AI-led)	0.000	0.115	0.251	0.021	0.032	0.003	0.017
Obs.	103	103	103	103	103	103	103

Note: SEs in parentheses; CIs in brackets. Human-only group omitted.

Controls: number of teammates; game-by-software fixed effects; maximum and minimum position skill fixed effects; attendance fixed effects.

*$P < 0.10$, **$P < 0.05$, and ***$P < 0.01$.

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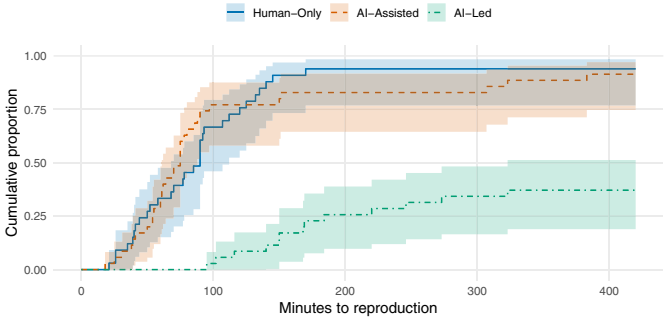


Fig. 1. Complementary Kaplan–Meier curves, showing the proportion of teams who computationally reproduced the paper by time *t* along with curve bands (95% CIs).

rate gap between human-only and AI-led teams was over 50 percentage points for most events in 2024. Of note, this gap had slightly narrowed by the final event of 2024.

2.2. Coding Errors or Data Irregularities. We have two primary dependent variables concerning coding error detection: counts of major and minor errors detected. We find that human-only teams identified on average 1.00 minor and 1.70 major errors, compared with 1.40 minor and 0.74 major errors for AI-assisted teams and 0.69 minor and 0.23 major errors for AI-led teams, respectively (Table 1). Table 2 provides OLS estimates indicating that, compared to AI-assisted and AI-led teams, human-only teams uncovered more major errors ($P = 0.006$ and $P < 0.001$, respectively). The difference in the number of minor coding errors detected is, however, not significant ($P = 0.421$ and $P = 0.347$, respectively). We further find that AI-assisted teams uncovered more minor errors than AI-led teams, but the estimate is not significant at any conventional level ($P = 0.115$). *SI Appendix* provides examples of errors and a discussion.

Fig. 2 plots complementary Kaplan–Meier curves showing how long teams took to find a first minor error (*Top* panel) and a first major error (*Bottom* panel). We find that the speed at which AI-assisted teams uncover a first (minor or major) error is not statistically significantly different from that of human-only teams and that AI-led teams are statistically significantly slower than human-only teams at uncovering a first major error.

Our findings suggest that human-only teams were more effective at detecting both major and minor errors compared to AI-led teams, highlighting a challenge in AI-led teams' ability to autonomously navigate and interpret complex code and detect data irregularities.

In exploratory analyses, we explore whether AI-led and AI-assisted teams are better at uncovering different types of errors, distinguishing between coding mistakes that require substantive understanding of the paper and those that do not. Table 3 provides OLS estimates indicating that, compared to AI-led teams, human-only teams uncovered more errors that occur in preparing the data and analysis (although not significantly different, $P = 0.165$), more errors related to the regression analysis ($P = 0.007$) and more transcription errors ($P = 0.034$). Human-only teams uncover more errors in these three categories than AI-assisted teams, but only one of the point estimates is statistically significant at the 10% level ($P = 0.993$, $P = 0.072$, and $P = 0.318$).

Our qualitative analysis (Section 2.5) suggests that some AI-led teams experienced prompt fatigue and hallucinated reasoning paths. This qualitative evidence motivates our exploratory analysis of whether AI-assisted and AI-led teams are more likely to produce false error detection. We find no evidence that this is the case ($P = 0.248$, $P = 0.642$), suggesting that hallucinations occur at other stages of the reproduction pipeline. This previous result may mask the fact that many AI-led teams did not detect any errors. We thus investigate the proportion of errors that remain undetected by each team. We find that AI-led teams detected a significantly smaller proportion of known errors than human-only and AI-assisted teams ($P < 0.001$, $P = 0.016$). These results suggest that AI-led teams' primary limitation lies in error discovery rather than erroneous overdetection.

We also provide noncausal evidence in exploratory analyses in *SI Appendix*, Table S14 that AI-assisted teams with more AI experience uncovered coding errors faster, although these estimates are only statistically significant at the 10% level ($P = 0.067$ and $P = 0.070$). Extending this analysis, *SI Appendix*, Table S15 compares human-only teams with AI-assisted teams with high vs. low/medium AI experience. These comparisons should be interpreted with caution, as the number of AI-assisted teams in each subgroup is small, resulting in limited statistical power. Nonetheless, the point estimates are consistent with the hypothesis that AI experience improves the effectiveness of AI-assisted teams. AI-assisted teams with high AI experience appear to uncover coding errors faster than human-only teams and detect more minor errors on average. The magnitudes of these differences are sizable, but the estimates are imprecise and not statistically significant at conventional levels. These findings are consistent with the behavioral evidence presented in Section 3.4, which examines how AI-assisted teams used ChatGPT.

We also find that Stata teams uncovered significantly more major errors ($P < 0.001$), with the human-only groups using Stata finding significantly more major errors than all other groups (*SI Appendix*, Table S13).

In an exploratory analysis, we investigate if the performance of AI-led teams in detecting errors improved over time. *SI Appendix*, Figs. S4 and S6 suggest no improvement of AI-led teams relative to human-only teams over the year 2024.

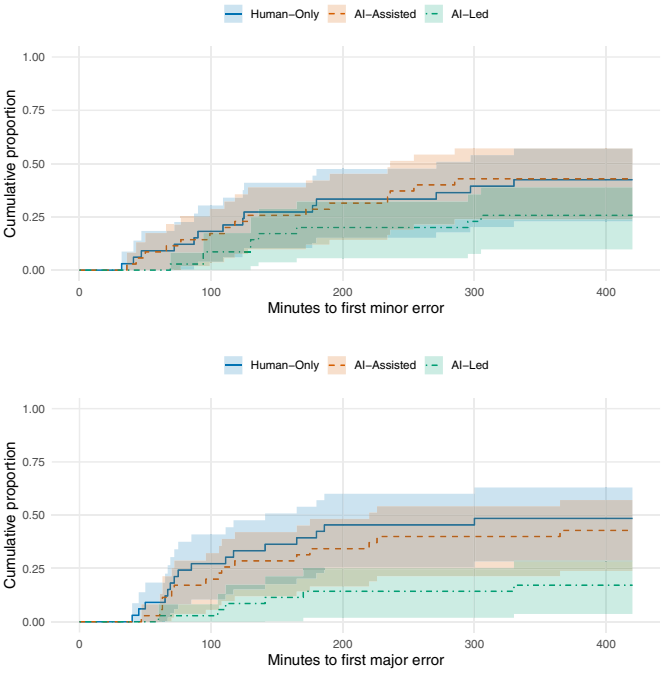


Fig. 2. Complementary Kaplan–Meier curves, showing the proportion of teams who found their first coding error by time *t* along with curve bands (95% CIs).

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Table 3. Causal relationship between treatment groups and error types

	(1) Preregression errors	(2) Regression errors	(3) Transcription/postregression errors	(4) False error detection	(5) Share of known errors not found
AI-assisted	-0.002	-0.604*	-0.343	-0.359	0.042
	(0.267)	(0.331)	(0.341)	(0.309)	(0.048)
	[-0.534; 0.530]	[-1.263; 0.055]	[-1.023; 0.337]	[-0.974; 0.256]	[-0.053; 0.137]
AI-led	-0.407	-0.886***	-0.652**	0.221	0.151***
	(0.290)	(0.321)	(0.301)	(0.473)	(0.043)
	[-0.986; 0.171]	[-1.524; -0.248]	[-1.251; -0.052]	[-0.721; 1.162]	[0.065; 0.236]
Controls	✓	✓	✓	✓	✓
Mean of dep. var	0.786	0.660	0.583	0.680	0.846
P-val (AI-assisted = AI-led)	0.140	0.228	0.274	0.146	0.016
Obs.	103	103	103	103	103

Note: SEs in parentheses; CIs in brackets. Human-only group omitted.

Controls: number of teammates; game-by-software fixed effects; maximum and minimum position skill fixed effects; attendance fixed effects.

*$P < 0.10$, **$P < 0.05$, and ***$P < 0.01$.

2.3. Proposed Robustness Checks. We find a clear, consistent performance hierarchy across both conditions: Human-only and AI-assisted teams outperform AI-led teams. We find that all human-only (33/33) and AI-assisted (35/35) teams proposed at least one good robustness check, whereas only 83% (29/35) of AI-led teams did so. Table 2 provides OLS estimates and show that the difference between AI-led groups and the other two groups is statistically significant ($P = 0.017$ and $P = 0.021$, respectively). We find that 29 of 33 human-only and 30 of 35 AI-assisted teams suggested two good checks, compared with just 22 of 35 AI-led teams (Table 2, $P = 0.022$ and $P = 0.032$).

Looking at whether teams report to have implemented those checks, AI-led teams were almost 32 percentage points less likely than the other two groups to report having conducted a robustness check that was classified as "good" ($P = 0.002$ and $P = 0.003$), and six AI-led teams supplied no robustness checks evaluated as "good" at all. These six teams' checks were judged as "bad" mostly because of a lack of clarity and duplicating analyses already run by the original authors.

Our results indicate that AI-led teams, while able to produce robustness checks with some level of quality, faced more challenges in aligning with the criteria. These difficulties may stem from omission of relevant information when describing the task to the AI or from limited ability of the AI to interpret the empirical strategy and to assess the feasibility of the checks.

2.4. Additional Analyses for AI-Assisted Teams. *SI Appendix*, Table S16 presents an exploratory correlational analysis examining the relationship between AI usage (measured by total prompts) and performance in AI-assisted teams. See *SI Appendix*, Fig. S7 for descriptive statistics on AI usage for AI-assisted teams. For this analysis, we divided teams into lower and higher AI-usage groups using a median split based on the total number of prompts they employed.

The findings indicate that AI-assisted teams with lower AI usage were less likely to achieve computational reproduction of the original results and uncovered less major and minor coding errors. Of note, our sample is small and none of the differences are statistically significant at the 5% level. To further explore potential mechanisms behind this heterogeneity, *SI Appendix*, Table S14 also reports differences in prompting behavior by AI experience. We find that AI-assisted teams with lower AI experience tend to interact with ChatGPT more frequently, as measured by the number of prompts, but the difference is not statistically significant due to the small sample size ($P = 0.307$). This pattern is consistent with the idea that less experienced teams rely more heavily on iterative prompting, which may contribute to longer task completion times. These results relate to a literature studying overreliance on AI support (20, 36, 37, 42–44).

2.5. Focus Groups. In additional exploratory analysis, between 18 April and 30 April 2025, we conducted six one-hour focus groups ($n = 25$) involving AI-led ($n = 8$), AI-assisted ($n = 11$), and human-only ($n = 6$) participants. The participants were aware of the headline quantitative results. While this creates risk of confirmation bias and demand characteristics, we addressed this by emphasizing process, task allocation, and failure points rather than outcomes in the discussion guide, and by treating focus group material as explanatory and triangulatory evidence rather than independent support for treatment differences. Accordingly, we use qualitative themes to illuminate mechanisms behind observed patterns, and we flag any tensions between participant claims and the experimental results. Consistent with this stance, where participant views exceeded what the quantitative results support, we report the discrepancy rather than treat it as confirmation.

Thematic analysis revealed the following patterns: AI-assisted participants reported that AI assistance sped things up, while AI-led participants reported the opposite. Human-only participants believed they were the most effective at detecting major errors, with AI-led participants trailing. AI-assisted teams strategically outsourced microtasks, for example boilerplate code and file location, while retaining conceptual control, whereas AI-led teams were required to cede entire analytic stages to ChatGPT and struggled when automation failed.

Data illuminated the practical consequences of these differences. Initial optimism about LLMs quickly gave way to prompt fatigue by participants, model's overconfidence, and mounting frustration, especially among AI-led participants who faced hallucinated paths, truncated context windows, and prolonged debugging loops. AI-assisted teams report that human expertise remained necessary for detecting subtle errors and for arbitrating disagreements between AI output and reality. Nevertheless, when used judiciously, LLMs accelerated routine work, suggested

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robustness checks, and broadened analytical ambition for less experienced coders. Therefore, our focus group findings imply that effective LLM prompting is becoming a specialized research skill and that near term gains will come from augmenting, not replacing, human judgment. Additional details on methodology and results from the focus group analysis are provided in *SI Appendix*.

3. Discussion

Computational reproducibility, error detection, and robustness checks are essential components of empirical research validation, but are resource-intensive tasks. Ensuring that research can be reproduced is financially demanding. Recent research suggests that, across 10 top economics journals, the average expense of reproducing a single study is about USD $365 (18). For the American Economic Association, data-editor activities cost approximately USD $750 per article (9). Against this backdrop, our comparative analysis of human-only, AI-assisted, and AI-led teams sheds light on how AI may be integrated into the costly reproducibility pipeline, potentially accelerating some stages of the process and reshaping how replication labor is allocated.

A key finding of our study is that AI-led teams were able to successfully computationally reproduce approximately 37% of results. This result is nontrivial and suggests that a first automated pass at computational reproducibility was already within reach for a meaningful subset of empirical work in 2024.

At the same time, our results temper expectations of immediate, widespread AI autonomy in reproducibility. While recent advances in large language models have expanded the scope for AI integration in research workflows (45, 46), AI-led and AI-assisted teams do not yet outperform human-only teams on average. Moreover, current AI deployments introduce additional costs, such as paid model subscriptions, without consistently delivering higher success rates. As a result, fully autonomous AI reproduction does not yet offer clear cost savings relative to experienced human researchers.

However, our study likely represents only a lower bound of the capabilities of a more fully developed autonomous AI replication system. In practice, an AI system could deploy more sophisticated prompting strategies, exploit parallel experimentation, and possibly be supervised by trained research assistants or undergraduate students rather than senior researchers, reducing labor costs while maintaining acceptable levels of oversight. Our findings thus imply that future iterations of AI-led reproducibility systems may achieve higher success rates without proportional increases in human effort.

This perspective reframes AI not as a replacement for human expertise, but as a tool for redistributing effort across stages of the reproducibility pipeline. AI systems may handle routine debugging, error detection, and preliminary robustness checks (47–49), while human researchers focus on interpretation, judgment, and more complex failures. Under this model, even partial automation can generate meaningful cost savings and efficiency gains at scale.

3.1. Summary of Findings. AI-led teams faced notable challenges compared to both AI-assisted and human-only teams. Only 37% of AI-led teams were able to successfully complete computational reproducibility, highlighting a substantial gap in the capacity of AI in 2024 to autonomously guide researchers through complex quantitative analyses. Similarly, in error detection, AI-led teams documented significantly fewer major and minor errors than either AI-assisted or human-only teams. These findings underscore the importance of still integrating human expertise. As LLMs continue to evolve, sustained benchmarking against humans will be crucial to ensure that future AI-led efforts close and potentially surpass the existing performance gap.

3.2. Limitations. One limitation is our sole focus on OpenAI's ChatGPT, meaning that we cannot generalize to all current AI models. Furthermore, the limited timeframe of seven hours for study teams to complete their reproductions may not adequately reflect the conditions under which reproducibility efforts are conducted depending on the field of science. In addition, participant incentives and attribution dynamics may have encouraged some teams to minimize time or effort, potentially increasing overreliance on AI tools. Finally, our analysis is based on a small, nonrandom set of studies spanning a limited range of social science methodologies and replication difficulty levels; although we provide detailed proxies for task complexity and error types, this sample composition constrains the extent to which our findings on AI assistance generalize across papers of different difficulty and across other scientific fields (*SI Appendix*, Table S5).

We note that participant behavior may have been influenced by observation and professional identity, generating a Hawthorne-type effect. Researchers with strong coding skills or a personal stake in reproducibility may have exerted greater effort in human-only teams, while responsibility may have been partially shifted to the AI in AI-assisted or AI-led settings. While this could bias relative performance comparisons, it may also reflect real-world incentive and attribution dynamics that shape how AI tools are adopted in research practice.

3.3. Implications for Human–AI Collaboration in Research. Our findings support the notion that, while AI tools hold promise for aiding in reproducibility tasks, the state of technology as of late 2024 is not yet advanced enough for full autonomy in complex empirical workflows. Human expertise remains critical to navigate challenges and provide interpretative guidance for reproducibility and error detection. The AI-assisted model—where humans work alongside AI tools—did not emerge as a winner over human-only teams in overall outcomes but outperformed AI-led teams on most of our outcomes.

In scenarios where computational reproducibility, error detection, and robustness checks require in-depth understanding, domain knowledge, and flexible problem-solving, human involvement currently adds value. The ability to contextualize, interpret, and implement complex quantitative research remains a human strength, highlighting the limits of current AI in fully autonomous reproduction.

3.4. Outlook. Advancements in models and further optimization of AI for reproduction may soon address the limitations we reported. Future advancements in models optimized through reinforcement learning to solve reasoning problems using chain of thought could address the limitations we reported, possibly improving the model's ability to reproduce complex quantitative research through iterative, reasoning-driven processes.

Future research should consider the potential for training models specifically in social science and quantitative research contexts. Current LLMs are trained on vast datasets but may lack specificity in understanding the unique demands of empirical social science research. AI systems tailored for social science reproduction (e.g., with native support for R and Stata) could potentially improve reproducibility outcomes, reducing the

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barriers AI currently faces in autonomously handling the nuances of quantitative research. Additionally, incorporating continuous feedback and learning mechanisms could allow AI-assisted and AI-led teams to improve performance over time, as AI learns from each reproduction task and adapts based on human feedback.

Future research should also focus on analyzing which prompting strategies leads to successful reproductions and which paths lead to failure, insights that could inform the development of AI systems better tailored for social science research. We make the chat transcripts publicly available and conduct an exploratory analysis of ChatGPT transcripts in *SI Appendix*.

4. Materials and Methods

Participants in the AI replication games experiments coauthor this study. The University of Ottawa Office of Research. Our preanalysis plan was preregistered on the Open Science Framework (OSF) on May 2nd, 2024, after our pilot event at the University of Toronto (https://osf.io/sz2g8/). AI-assisted and AI-led teams took part in a mandatory one-hour ChatGPT training, while the same training was optional for human-only teams; slides and recordings are available on OSF. A version-tagged copy of the code and data is permanently archived at https://github.com/I4Replication/AI-Games, and we make our AI training materials and recording, data and code, preanalysis plan, and template form available at https://osf.io/sz2g8/ with no restrictions on sharing or reuse.

5. Research Ethics Boards

Participants in the AI replication games experiments coauthor this study. The University of Ottawa Office of Research Ethics and Integrity reviewed and approved our AI games (H-09-25-12041). The King's College London Research Ethics Office reviewed and approved our focus groups (MRA-24/25-48393). All participants provided informed consent.

Data, Materials, and Software Availability. We make our i) AI training materials and recording, ii) data and code, iii) preanalysis plan and iv) template form available here: https://github.com/I4Replication/AI-Games (50). We declare no restrictions on sharing or reuse.

ACKNOWLEDGMENTS. We would like to thank Gabriel Zimmerman for research assistance. This research and AI replication games were funded by Coefficient Giving project "Benchmarking LLM agents on real-world tasks: Reproducibility" and the Alfred P. Sloan Foundation Foundation grant G-2023-22326. We also benefited from funding to host games from the Universities of Toronto, Ottawa, Cornell, and Tilburg. Mahmoud Elsherif acknowledges funding from Leverhulme Early Career Research Fellowship-ECF-2022-761. S. Akhtar acknowledges funding DP200102935 awarded by the Australian Research Council Grant. R. Seri acknowledges funding from project "Dipartimento di Eccellenza 2023-2027" awarded by the Ministero dell'Università e della Ricerca.

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